CSCI 112 Assignment – Arrays and Loops

Points: 10

Automated tests will run on GitHub Actions after every push. Check the Actions tab to see which tests pass and which fail.

Instructions

1. **Fork** the repository on GitHub - `https://github.com/jpach-cs/112-ass05-template`.
2. Clone your forked repository to your local machine.
3. Open the file `code.h` and add:
    1. Include guards at the top: `#ifndef CODE_H` `#define CODE_H`
    2. All function prototypes
    3. Include guards at the bottom: `#endif`
4. Open the file `code.c`.
    1. Implement every function listed below:
        - Read the comments above each function carefully.
        - Follow the rules — use ONLY `if`, `switch`, `while`, `for`, `putchar()`, `getchar()`, `printf()`, `sprintf()`.
        - Do NOT use `goto`, or any standard library functions.
    2. Test your implementation locally using `main.c`.
5. Once all functions are complete — save your changes, then **commit** and **push** to your repository.
6. Go to the **Actions** tab on GitHub and check the test results.

Learning Objectives

After completing this lab, you will:

Use `for` loops to iterate over arrays with known length Use `while` loops to search and converge on solutions Implement array manipulation functions (`min`, `max`, `sum`, `average`) Implement search algorithms (`linear_search`) Implement iterative mathematical algorithms (`heron` - Heron's method) Work with multi-file C projects using header files with include guards.

Implement 6 functions that work with arrays and use iteration:

Function	Purpose	Loop Type	Return Type
`find_max()`	Find largest element	`for`	`int`
`find_min()`	Find smallest element	`for`	`int`
`sum_array()`	Sum all elements	`for`	`long`

Function	Purpose	Loop Type	Return Type
`average()`	Calculate mean	`for`	`double`
`linear_search()`	Find element index	`while`	`int`
`heron()`	Calculate square root	`while`	`double`

Background

For Loops vs While Loops

Use `for` loops when you know the iteration count:

```
for (int i = 0; i < n; i++)
{
    // Process array[i]
}
```

Use `while` loops when you don't know when to stop:

```
int i = 0;
while (i < n)
{
    if (/* found it */)
        return i;
    i++;
}
```

Include Guards

Header files must use include guards to prevent multiple inclusion:

```
#ifndef CODE_H
#define CODE_H

int find_max(int arr[], int n);
// ... more prototypes

#endif
```

Heron's Method for Square Root

Background:

Heron's method iteratively improves an initial guess:

```
next_guess = (current_guess + x / current_guess) / 2.0
```

Stop when consecutive guesses differ by less than epsilon.

Algorithm:

1. Start with initial guess: guess = x / 2.0
2. Use while (1) loop (infinite loop with break):
   - Save current guess: prev_guess = guess
   - Calculate next guess: guess = (guess + x / guess) / 2.0
   - Calculate difference: diff = guess - prev_guess
   - Make difference positive manually: if (diff < 0) diff = -diff;
   - If diff < epsilon, break the loop
3. Return the final guess

Note: You cannot use fabs() - manually check if negative with if (x < 0) x = -x;

Example: heron(4.0, 0.0001); // Returns 2.0 heron(9.0, 0.0001); // Returns 3.0 heron(2.0, 0.0001); // Returns 1.414

#Header comment — required in code.c

```
// Your Name
// CSCI 112 Fall 2026
// Programming Assignment #1
// I declare that I am the author of this work, take full responsibility for it, and have d
```

Only for testing

Local Testing (Sandbox)

Create test cases in main.c:

```c
#include <stdio.h>
#include "code.h"

int main()
{
    int arr[] = {5, 2, 8, 1, 9};
    printf("Max: %d\n", find_max(arr, 5));
    printf("Min: %d\n", find_min(arr, 5));
    printf("Sum: %ld\n", sum_array(arr, 5));

    float farr[] = {2.0f, 4.0f, 6.0f};
    printf("Average: %f\n", average(farr, 3));
```

```
    printf("Search for 8: %d\n", linear_search(arr, 5, 8));
    printf("Search for 7: %d\n", linear_search(arr, 5, 7));

    printf("Square root of 4: %f\n", heron(4.0, 0.0001));
    printf("Square root of 2: %f\n", heron(2.0, 0.0001));

    return 0;
}
```